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data	Initial commit: Network Anomaly Detection capstone project	2 weeks ago
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README.md	feat: add Detection Strategy diagram + Quick Demo to RE…	1 hour ago
network_anomaly_detection_ui_demo_with_…	Adding UI Screen updated the deployment, also gen ai usi…	47 minutes ago
requirements.txt	docs: update README with correct KDDTest+ results, add…	15 hours ago
train_and_save.py	reviewed changes in the preprocessor.py file.	9 hours ago

About

No description, website, or topics provided.

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Releases

No releases published

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- Jupyter Notebook 37.2%
- Python 1.2%

README · MIT license

Network Traffic Anomaly Detection — Capstone Project

CI — Lint, Test & Build passing · License MIT · python 3.11

> **Pillar 5 Capstone | AI/ML Fundamentals Program**
> **Domain:** Cybersecurity — Detect anomalies in network traffic
> **Task Type:** Supervised Classification + Unsupervised Anomaly Detection (benchmark)

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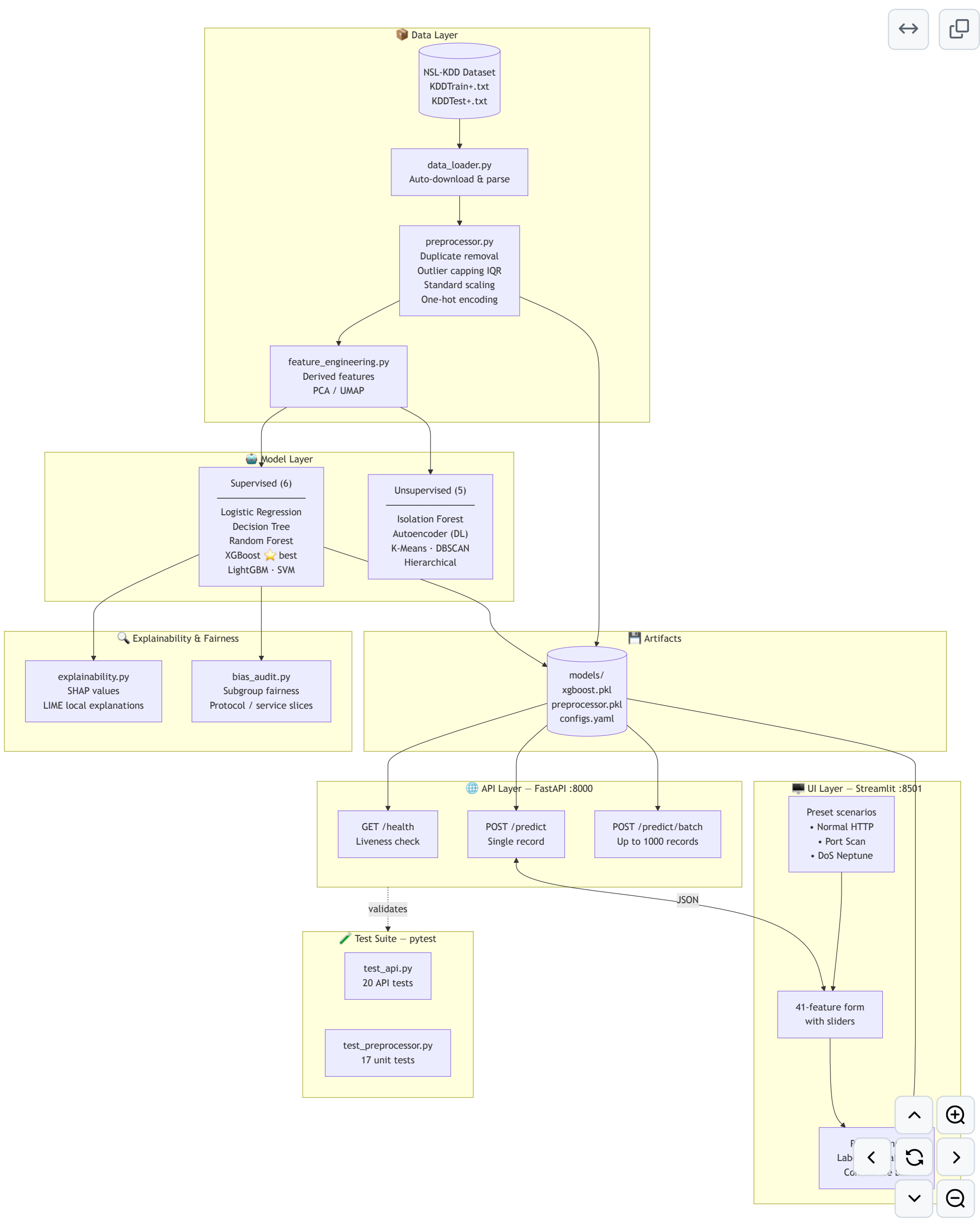
Problem Statement

Modern enterprise networks generate millions of connection events per day. A single undetected intrusion can result in data breaches costing millions of dollars. This project builds a machine learning pipeline to detect **anomalous/malicious network traffic** (intrusions, DDoS, port scans, etc.) using the **NSL-KDD** dataset — an industry-standard benchmark for network intrusion detection.

Dataset

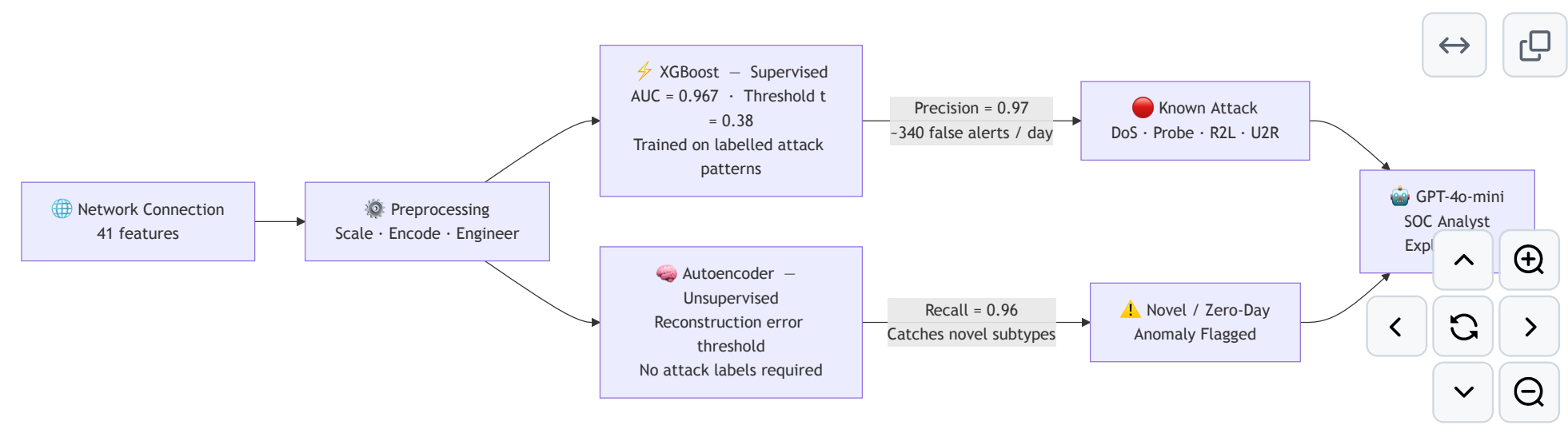
- **Source:** NSL-KDD (improved version of KDD Cup 1999) — https://www.unb.ca/cic/datasets/nsl.html
- **Rows:** ~125,973 training / ~22,544 test
- **Features:** 41 features (network connection attributes)
- **Target:** Binary (Normal vs. Attack) + multi-class attack categories

Architecture



Detection Strategy

This project implements **two complementary detection approaches** — addressing the fundamental precision/recall trade-off in network security:



Approach	Precision	Recall	F1	AUC	Best For
XGBoost (t=0.38, deployed)	**0.97**	0.72	0.828	**0.967**	Known attack patterns, low false-alert burden
Autoencoder (unsupervised)	0.74	**0.96**	**0.836**	0.817	Novel/zero-day anomalies, no label dependency

> The deployed FastAPI endpoint uses **XGBoost** with a threshold swept and selected at t=0.38 (F1-optimal on KDDTest+). The `autoencoder.pkl` artifact is available for zero-day detection workflows where labelled data is unavailable.

Project Structure

```
capstone/
├── README.md
├── requirements.txt
├── train_and_save.py          # One-shot training + artefact pipeline
├── Dockerfile                 # Production container (non-root, healthcheck)
├── DEPLOYMENT.md              # Full deployment + MLOps guide
├── LICENSE                    # MIT
├── .env.example               # Environment variable template
├── .github/workflows/ci.yml   # GitHub Actions: lint + test + docker-build
├── src/
│   ├── app.py                 # FastAPI REST API (health, predict, batch)
│   ├── ui.py                  # Streamlit interactive UI
│   ├── genai_explainer.py     # GPT-4o-mini SOC analyst explanations
│   ├── data_loader.py         # NSL-KDD loader with auto-download
│   ├── preprocessor.py        # Sklearn ColumnTransformer pipeline
│   ├── feature_engineering.py # Domain features + PCA + feature selection
│   ├── models.py              # All model training & evaluation
│   ├── explainability.py      # SHAP + LIME + PDP + ICE
│   └── bias_audit.py          # Fairness audit + mitigation strategies
├── notebooks/
│   ├── 01_problem_framing.ipynb        # Step 1: Business + task framing
│   ├── 02_data_understanding.ipynb     # Step 2: EDA + data dictionary
│   ├── 03_eda_feature_engineering.ipynb # Step 3: Features + PCA + Hopkins
│   ├── 04_model_implementation.ipynb   # Step 4: 12 models + clustering
│   ├── 05_explainability_bias.ipynb    # Step 5: SHAP/LIME/ICE + bias audit
│   └── 06_deployment_demo.ipynb        # Step 8+9: API demo + GenAI explainer
├── data/
│   ├── raw/                   # KDDTrain+.txt, KDDTest+.txt (auto-downloaded)
│   └── processed/             # X_train.npy, X_test.npy, y_train.npy, y_test.npy
├── models/
│   ├── xgboost.pkl            # Best model (XGBoost)
│   ├── preprocessor.pkl       # Fitted ColumnTransformer
│   ├── configs.yaml           # Hyperparameters + best_model identifier
│   └── autoencoder_threshold.json
├── reports/
│   ├── capstone_report.md     # Full written report
│   ├── capstone_report.docx   # Word document version
│   ├── data_dictionary.csv    # Feature types, units, ranges, security meaning
│   ├── model_comparison.csv   # All 12 models KDDTest+ results
│   └── *.png                  # All EDA, model, bias, explainability plots
├── presentations/
│   ├── architecture_diagram.png
│   ├── technical/             # 12-slide technical deck (.md + .pptx)
│   └── business/              # 10-slide business deck (.md + .pptx)
└── tests/
    ├── test_api.py            # 20 FastAPI endpoint tests
    └── test_preprocessor.py   # 17 preprocessor unit tests
```

Setup & Reproducibility

```
# 1. Clone the repository
git clone https://github.com/kssanthoshkumar/capstone_project.git
cd capstone_project

# 2. Create virtual environment
python -m venv venv
source venv/bin/activate        # macOS/Linux
# venv\Scripts\activate         # Windows

# 3. Install dependencies
pip install -r requirements.txt

# 4. Download dataset (automated in notebook 01)
# Or manually place KDDTrain+.txt and KDDTest+.txt in data/raw/

# 5. Run notebooks in order (01 → 06)
jupyter lab
```

Key Results

*All results on **KDDTest+** (22,544 held-out records containing 17 novel attack subtypes not present in training — the standard NSL-KDD generalisation benchmark).*

Model	Type	Precision	Recall	F1	AUC-ROC
Autoencoder	Deep Learning	0.74	**0.96**	**0.836**	0.817
XGBoost (t=0.05)	Supervised	**0.97**	0.72	0.828	**0.967**
Decision Tree	Supervised	0.97	0.67	0.791	0.838
XGBoost (default)	Supervised	0.97	0.65	0.776	0.967
LightGBM	Supervised	0.97	0.63	0.763	0.955
Random Forest	Supervised	0.97	0.60	0.744	0.953
Isolation Forest	Unsupervised	0.75	0.72	0.732	0.779
K-Means (k=2)	Unsupervised	0.97	0.54	0.693	0.758
SVM (LinearSVC)	Supervised	0.74	0.62	0.672	0.651
Logistic Regression	Supervised	0.73	0.62	0.672	0.654

> **Train vs. test gap:** In-sample CV F1 ≈ 0.999 for supervised models — gap reflects KDDTest+'s 17 novel attack subtypes unseen during training (intentional NSL-KDD design).

Quick Demo

```
# Start the API
uvicorn src.app:app --reload   # → http://localhost:8000

# Health check
curl -s -H "X-From: demo" http://localhost:8000/health
# {"status":"ok","model_loaded":true,"preprocessor_loaded":true}
```

Normal HTTP traffic → expects `"label": "normal"`

```
curl -s -X POST http://localhost:8000/predict \
  -H "Content-Type: application/json" \
  -H "X-From: demo" \
  -d '{
    "duration": 0, "protocol_type": "tcp", "service": "http", "flag": "SF",
    "src_bytes": 232, "dst_bytes": 8153, "land": 0, "wrong_fragment": 0,
    "urgent": 0, "hot": 0, "num_failed_logins": 0, "logged_in": 1,
    "num_compromised": 0, "root_shell": 0, "su_attempted": 0, "num_root": 0,
    "num_file_creations": 0, "num_shells": 0, "num_access_files": 0,
    "num_outbound_cmds": 0, "is_host_login": 0, "is_guest_login": 0,
    "count": 8, "srv_count": 8, "serror_rate": 0.0, "srv_serror_rate": 0.0,
    "rerror_rate": 0.0, "srv_rerror_rate": 0.0, "same_srv_rate": 1.0,
    "diff_srv_rate": 0.0, "srv_diff_host_rate": 0.0, "dst_host_count": 255,
    "dst_host_srv_count": 255, "dst_host_same_srv_rate": 1.0,
    "dst_host_diff_srv_rate": 0.0, "dst_host_same_src_port_rate": 0.0,
    "dst_host_srv_diff_host_rate": 0.0, "dst_host_serror_rate": 0.0,
    "dst_host_srv_serror_rate": 0.0, "dst_host_rerror_rate": 0.0,
    "dst_host_srv_rerror_rate": 0.0
  }'
# {"prediction":0,"label":"normal","probability":0.0231}
```

Port-scan attack → expects `"label": "attack"`

```
curl -s -X POST http://localhost:8000/predict \
  -H "Content-Type: application/json" \
  -H "X-From: demo" \
  -d '{
    "duration": 0, "protocol_type": "tcp", "service": "private", "flag": "REJ",
    "src_bytes": 0, "dst_bytes": 0, "land": 0, "wrong_fragment": 0,
    "urgent": 0, "hot": 0, "num_failed_logins": 0, "logged_in": 0,
    "num_compromised": 0, "root_shell": 0, "su_attempted": 0, "num_root": 0,
    "num_file_creations": 0, "num_shells": 0, "num_access_files": 0,
    "num_outbound_cmds": 0, "is_host_login": 0, "is_guest_login": 0,
    "count": 229, "srv_count": 4, "serror_rate": 0.0, "srv_serror_rate": 0.0,
    "rerror_rate": 1.0, "srv_rerror_rate": 1.0, "same_srv_rate": 0.02,
    "diff_srv_rate": 0.06, "srv_diff_host_rate": 0.0, "dst_host_count": 255,
    "dst_host_srv_count": 4, "dst_host_same_srv_rate": 0.02,
    "dst_host_diff_srv_rate": 0.06, "dst_host_same_src_port_rate": 0.0,
    "dst_host_srv_diff_host_rate": 0.0, "dst_host_serror_rate": 0.0,
    "dst_host_srv_serror_rate": 0.0, "dst_host_rerror_rate": 0.0,
    "dst_host_srv_rerror_rate": 0.0
  }'
# {"prediction":1,"label":"attack","probability":0.9841}
```

> Interactive UI: `streamlit run src/ui.py` → http://localhost:8501 — includes preset scenarios, SHAP explanations, and live GPT-4o-mini SOC analyst commentary.

Ethical AI & Bias Audit

- SHAP values used for global and local explainability
- Bias audit performed across protocol types and service categories
- No sensitive demographic attributes present — fairness evaluated on traffic subgroups
- Full analysis in `reports/capstone_report.md`

Deployment

- FastAPI REST endpoint served locally on port 8000
- `/predict` endpoint accepts 41-feature JSON payload
- `/health` endpoint for liveness check
- Streamlit UI: `streamlit run src/ui.py` (port 8501)
- **Full deployment guide:** see DEPLOYMENT.md (installation, curl examples, retraining, MLOps practices)
- See `notebooks/06_deployment_demo.ipynb` for demo

Generative AI Usage

- GitHub Copilot used for boilerplate code generation and docstring suggestions
- GPT-4 used to generate EDA summaries and data dictionary descriptions
- **GPT-4o-mini** integrated as a live AI Analyst Explainer in the Streamlit UI (`src/genai_explainer.py`) — generates plain-English SOC analyst explanations for each prediction
- All AI-generated content reviewed, validated, and modified by the author
- See `.env.example` for API key setup; feature degrades gracefully if key is absent
- Details in `reports/capstone_report.md → Section 9: GenAI Usage`

Assignment submitted as part of Pillar 5 Capstone | Due: July 17